

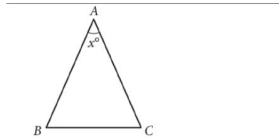
SAT Math

Lines, Angles, and Triangles 1

Question # ID
1.1 c8d60e48

Answer

B

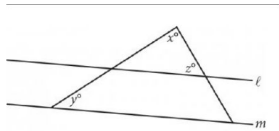


In the given triangle, $AB = AC$ and $\angle ABC$ has a measure of 67° . What is the value of x ?

- A. 36
- B. 46
- C. 58
- D. 70

1.2 a6dbad6b

B



Note: Figure not drawn to scale.

In the figure above, lines l and m are parallel, $y = 20$, and $z = 60$. What is the value of x ?

- A. 120
- B. 100
- C. 90
- D. 80

1.3 cbe8ca31

A

In $\triangle XYZ$, the measure of $\angle X$ is 24° and the measure of $\angle Y$ is 98° . What is the measure of $\angle Z$?

- A. 58°
- B. 74°
- C. 122°
- D. 212°

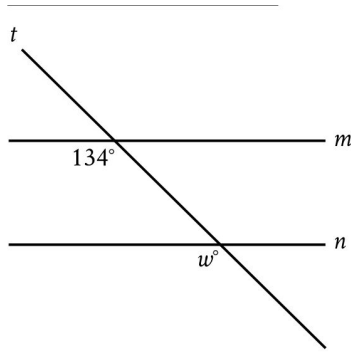
SAT Math

Lines, Angles, and Triangles 1

Question # ID
1.4 c24e1bda

Answer

D



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n . What is the value of w ?

- A. 13
- B. 34
- C. 66
- D. 134

1.5 3563d76d

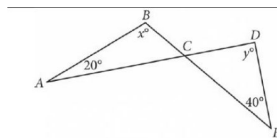
C

At a certain time and day, the Washington Monument in Washington, DC, casts a shadow that is 300 feet long. At the same time, a nearby cherry tree casts a shadow that is 16 feet long. Given that the Washington Monument is approximately 555 feet tall, which of the following is closest to the height, in feet, of the cherry tree?

- A. 10
- B. 20
- C. 30
- D. 35

1.6 dfc420b2

C



Note: Figure not drawn to scale.

In the figure above, $\overline{AD}$ intersects $\overline{BE}$ at C. If

$x = 100$, what is the value of y ?

- A. 100
- B. 90
- C. 80
- D. 60

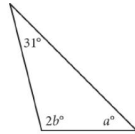
SAT Math

Lines, Angles, and Triangles 1

Question # ID
1.7 410bdb66

Answer

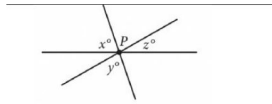
A



In the triangle above, $a = 45$. What is the value of b ?

- A. 52
- B. 59
- C. 76
- D. 104

1.8 087cdcfd



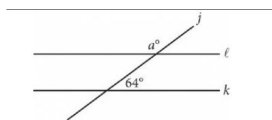
Note: Figure not drawn to scale.

In the figure, three lines intersect at point P . If $x = 65$ and $y = 75$, what is the value of z ?

- A. 140
- B. 80
- C. 40
- D. 20

C

1.9 992f4e93



Note: Figure not drawn to scale.

In the figure above, lines ℓ and k are parallel. What is the value of a ?

- A. 26
- B. 64
- C. 116
- D. 154

C

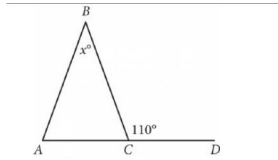
SAT Math

Lines, Angles, and Triangles 1

Question # ID
1.10 5733ce30

Answer

D

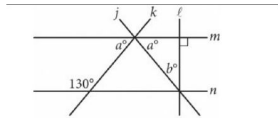


In the given figure, $\overline{AC}$ extends to point D . If the measure of $\angle BAC$ is equal to the measure of $\angle BCA$, what is the value of x ?

- A. 110
- B. 70
- C. 55
- D. 40

1.11 3828f53d

A



Note: Figure not drawn to scale.

In the figure above, lines m and n are parallel. What is the value of b ?

- A. 40
- B. 50
- C. 65
- D. 80

1.12 42b4493b

B

In a right triangle, the measure of one of the acute angles is 51° . What is the measure, in degrees, of the other acute angle?

- A. 6
- B. 39
- C. 49
- D. 51

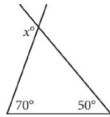
SAT Math

Lines, Angles, and Triangles 1

Question # ID
1.13 36200a38

Answer

B

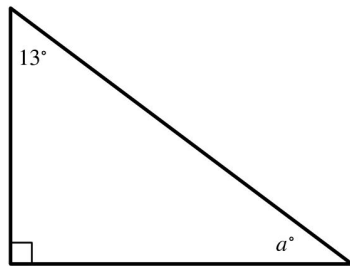


In the figure above, two sides of a triangle are extended. What is the value of x ?

- A. 110
- B. 120
- C. 130
- D. 140

1.14 69f4bbdc

B



Note: Figure not drawn to scale.

In the right triangle shown, what is the value of a ?

- A. 13
- B. 77
- C. 90
- D. 103

1.15 f1747a6a

D

In triangle ABC , the measure of angle B is 52° and the measure of angle C is 17° . What is the measure of angle A ?

- A. 21°
- B. 35°
- C. 69°
- D. 111°

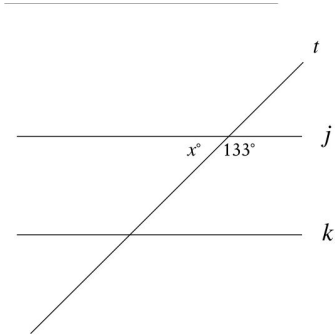
SAT Math

Lines, Angles, and Triangles 1

Question # ID
1.16 3b4b5b1e

Answer

47

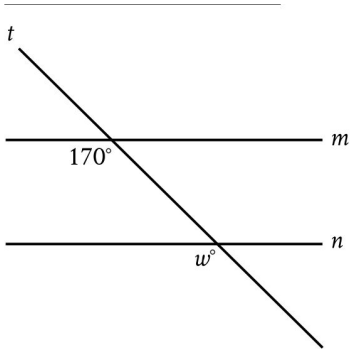


Note: Figure not drawn to scale.

In the figure, line j is parallel to line k . What is the value of x ?

1.17 5207e508

D



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n . What is the value of w ?

- A. 17
- B. 30
- C. 70
- D. 170

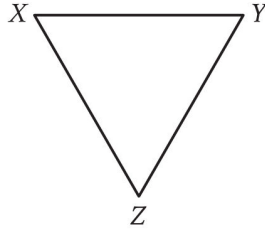
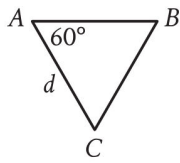
SAT Math

Lines, Angles, and Triangles 1

Question # ID
1.18 e0d2e21a

Answer

C



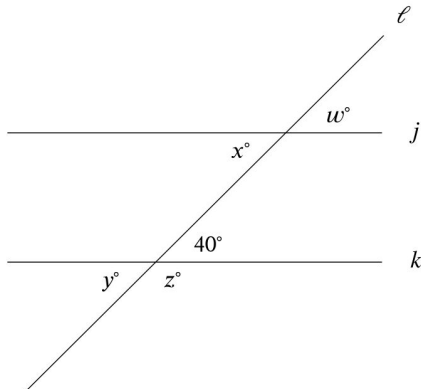
Note: Figures not drawn to scale.

For the triangles shown, triangle ABC is dilated by a scale factor of 3 to obtain triangle XYZ , where $d = 16$. What is the measure, in degrees, of angle X ?

- A. 20
- B. 57
- C. 60
- D. 63

1.19 9d078710

A



Note: Figure not drawn to scale.

In the figure shown, line ℓ intersects lines j and k . Which additional piece of information is sufficient to prove that lines j and k are parallel?

- A. $w = 40$
- B. $x = 140$
- C. $y = 40$
- D. $z = 140$